

Elsa Elisa Yohana Sianturi

Bandar Lampung, Indonesia | +62-858-3446-5676 | elsaelisayohana05@gmail.com | LinkedIn: [linkedin.com/in/elsaelisayohanasanturi](https://www.linkedin.com/in/elsaelisayohanasanturi) | GitHub: github.com/elsaelisa09 | Portfolio: elsaelisayohanasanturi.vercel.app

Fresh Graduate in Informatics Engineering with a focus on AI Engineering, Machine Learning, NLP, and Computer Vision. Skilled in PyTorch, Hugging Face, Scikit-learn, OpenCV, and Transformer-based architectures through projects involving multimodal CLIP + ELECTRA, BM25-based RAG systems, IndoBERT for NER, and EfficientNet image classification. Backed by hands-on industry experience at PT Pertamina Hulu Rokan.

EDUCATION

B.Eng. in Informatics Engineering - Institut Teknologi Sumatera
Lampung, Indonesia

Aug 2022 - May 2026

- GPA: 3.64/4.00
- Focus: AI Engineering, Machine Learning, NLP, and Computer Vision.
- Relevant coursework: Machine Learning, Deep Learning, NLP, Multimodal Machine Learning.

EXPERIENCE

AI Engineer Intern — PT Pertamina Hulu Rokan
Rumbai, Pekanbaru, Indonesia

Jun 2025 - Jul 2025

- Developed Retrieval-Augmented Generation (RAG) and LLM-based system for question answering over internal documents using Python and PyTorch.
- Applied NLP techniques and built document retrieval pipelines to improve information access and support AI-driven document digitalization initiatives.

Teaching Assistant — UPA TIK ITERA
Lampung, Indonesia

Aug 2023 - Present

- **Introduction to Computing & Software I (PKS I):** Guided students in procedural programming and fundamental logic. (Credential ID: B/5359/IT9.E2/TA.00.00/2024)
- **Interaction Design:** Provided technical mentoring for UI/UX project development and user-centered design principles. (Credential ID: 33133/IT9.3.3/TA.00.00/2025)
- **Operating Systems:** Facilitated hands-on laboratory sessions focusing on process management and memory allocation.

PROJECTS

Multimodal Deep Learning for Self-Harm Meme Detection | [GitHub]
Tech: CLIP, ELECTRA, Gradio

Aug 2025 - Apr 2026

- Developed multimodal deep learning system combining CLIP and ELECTRA for automatic self-harm meme detection with 96% accuracy.
- Deployed inference system using Gradio on Hugging Face Spaces.

Multimodal City Branding Analysis | [GitHub]
Tech: Image-Text Representation Learning, Social Media Analysis

May 2026 - Present

- Developing multimodal AI system for tourism analysis using Instagram image-text data classification.
- Collaborating with Urban and Regional Planning program on AI modelling research.

Document AI ChatBot with RAG & LLM | [GitHub]
Tech: RAG, LLM Integration, Semantic Search

Jun 2025 - Jul 2025

- Built Document AI chatbot integrating Retrieval-Augmented Generation with LLMs for intelligent document question answering.
- Developed retrieval and contextual response generation pipelines for semantic search.

LEADERSHIP & ORGANIZATIONS

Head of Academic & Scholarship Division — HMIF ITERA

Jan 2025 – Dec 2025

Lampung, Indonesia

- Managing academic programs and scholarship opportunities for Informatics Engineering students.

Head of Entrepreneurship Department — Sobat Bumi Lampung

Oct 2025 – Present

Lampung, Indonesia

- Leading entrepreneurial initiatives and community projects focused on sustainability and environmental awareness.

Secretary of Minister of Information Technology — KM ITERA

Apr 2024 – Dec 2024

Lampung, Indonesia

- Managed administrative documentation and coordinated IT-related projects within the student government.

Staff of Publication & Documentation Division — HMIF ITERA

Mar 2024 – Dec 2024

Lampung, Indonesia

- Documented organizational events and managed public relations materials for the Informatics Student Association.

SKILLS

- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, LLMs, RAG, Fine-Tuning
- **AI Development:** Multimodal Learning, NER, Image Classification, Audio Classification
- **Frameworks & Tools:** PyTorch, Scikit-learn, Hugging Face, OpenCV, Transformers, W&B
- **Programming:** Python, SQL, Git, Linux
- **Others:** Microsoft Power BI, Figma

CERTIFICATIONS

- **AWS re/Start Cloud Computing Program** - Orbit Future Academy & Amazon Web Services (2025 - Present)
- **Belajar Pemrograman Prosedural dengan Python** - Dicoding (2024) | Credential ID: 2VX3O4W23ZYQ
- **Memulai Pemrograman dengan Python** - Dicoding (2024) | Credential ID: L4PQQ8O2PO01
- **Belajar Dasar Structured Query Language (SQL)** - Dicoding (2024) | Credential ID: NVP77M3WVPR0
- **AI Praktis untuk Productivity** - Dicoding (2024) | Credential ID: 1OP82QOYLPQK

AWARDS & HONORS

- **Pertamina Sobat Bumi Scholarship Recipient** - Pertamina Foundation (2024)
- **2nd Winner of UI/UX Competition Point Project 2.0** - HMIF ITERA (2024) | Credential ID: 313/525/SRT/HMIF/XII/2024

LANGUAGES & PUBLICATIONS

- **Languages:** Indonesian (Native), English (Professional Working Proficiency)
- **Publications:** Sianturi, E. E. Y., et al. "Transformasi Digital Layanan Posyandu." PADIMAS, 2025. | [Full Paper]